Portfolio 3: Transactions and Concurrency

Implementing **START TRANSACTION, COMMIT,** and **ROLLBACK.**

```
USE UniversityManagementDB;

-- Check Mary's current score
SELECT * FROM Enrollment
WHERE enrollment_id = 3;

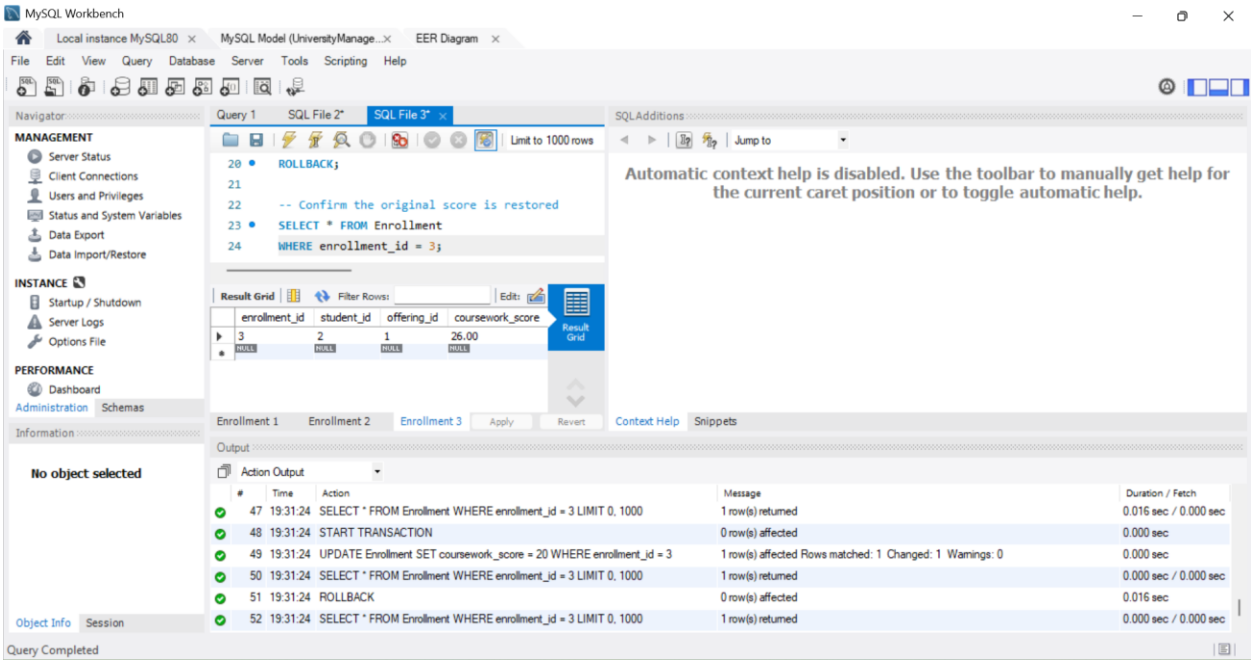
-- Begin a transaction
START TRANSACTION;

-- Temporarily change Mary's coursework score
UPDATE Enrollment
SET coursework_score = 20
WHERE enrollment_id = 3;

-- Check the temporary change
SELECT * FROM Enrollment
WHERE enrollment_id = 3;

-- Undo the change
ROLLBACK;

-- Confirm the original score is restored
SELECT * FROM Enrollment
WHERE enrollment_id = 3;
```




```
USE UniversityManagementDB;

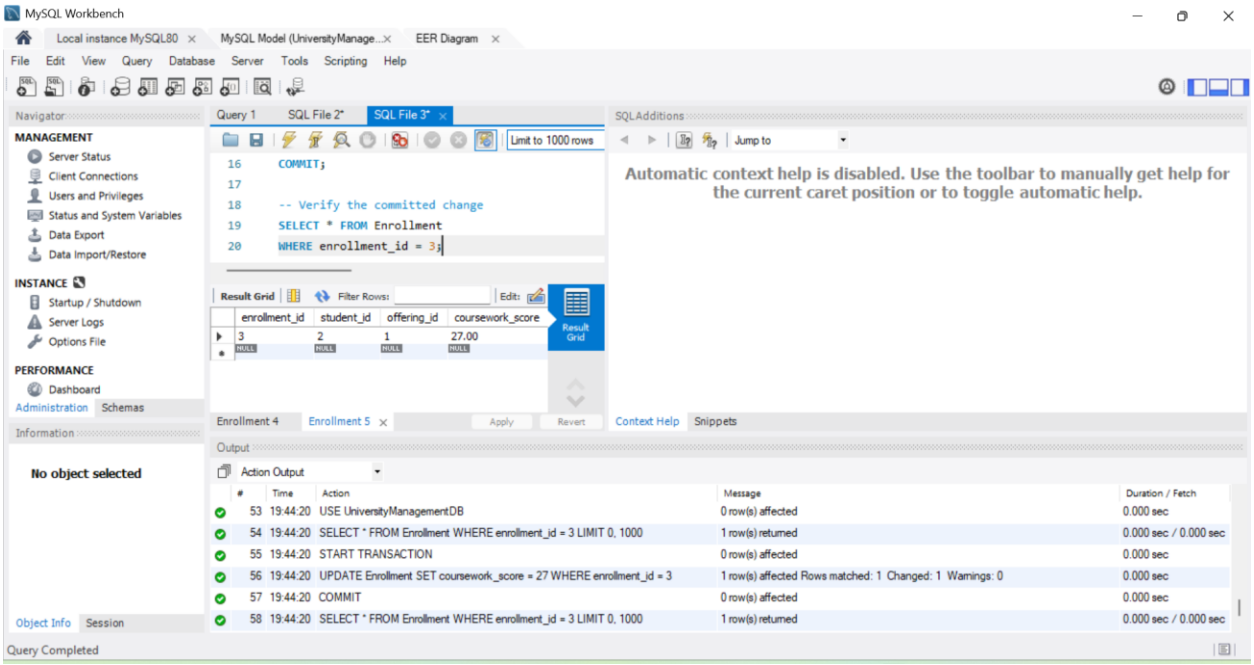
-- Check the original record
SELECT * FROM Enrollment
WHERE enrollment_id = 3;

-- Begin transaction
START TRANSACTION;

-- Update Mary's coursework score
UPDATE Enrollment
SET coursework_score = 27
WHERE enrollment_id = 3;

-- Make the change permanent
COMMIT;

-- Verify the committed change
SELECT * FROM Enrollment
WHERE enrollment_id = 3;
```



Transaction Isolation

USE UniversityManagementDB;

SET SESSION TRANSACTION ISOLATION LEVEL REPEATABLE READ;

START TRANSACTION;

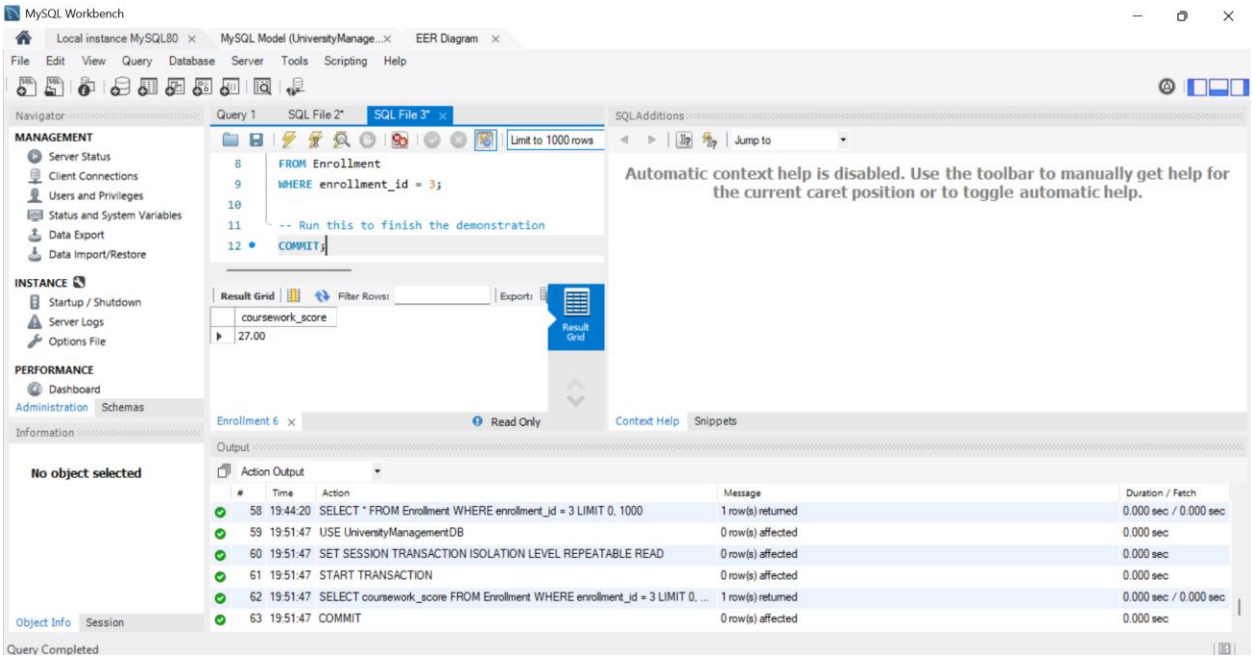
SELECT coursework_score

FROM Enrollment

WHERE enrollment_id = 3;

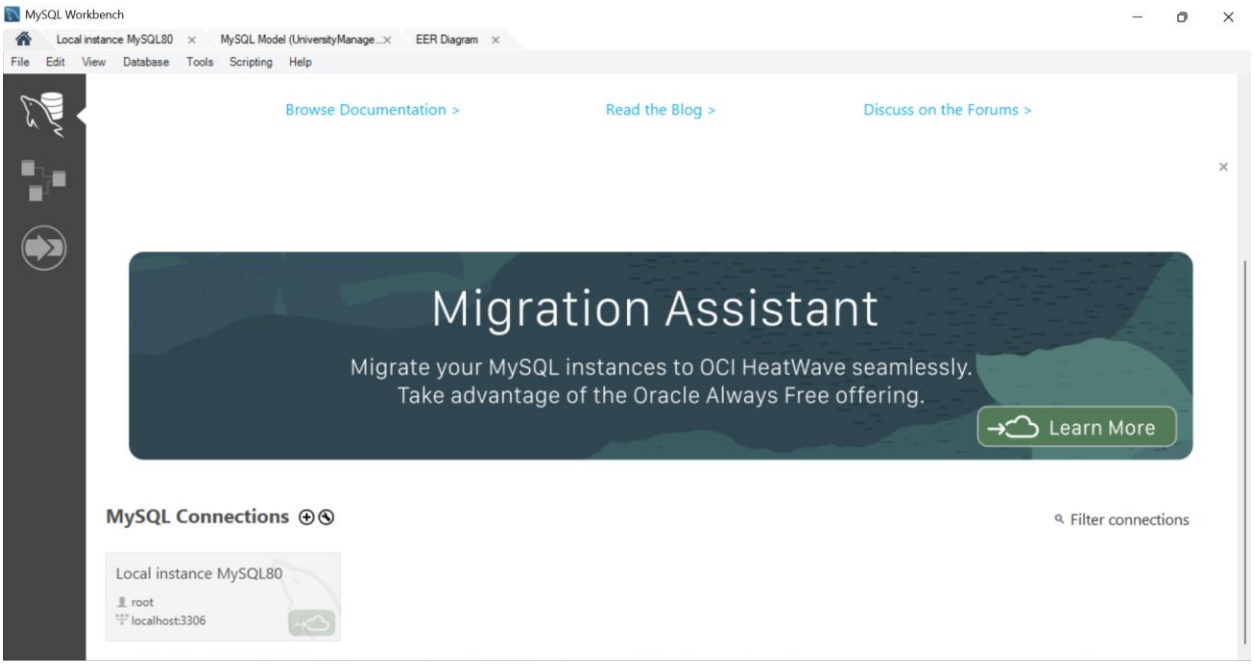
-- Run this to finish the demonstration

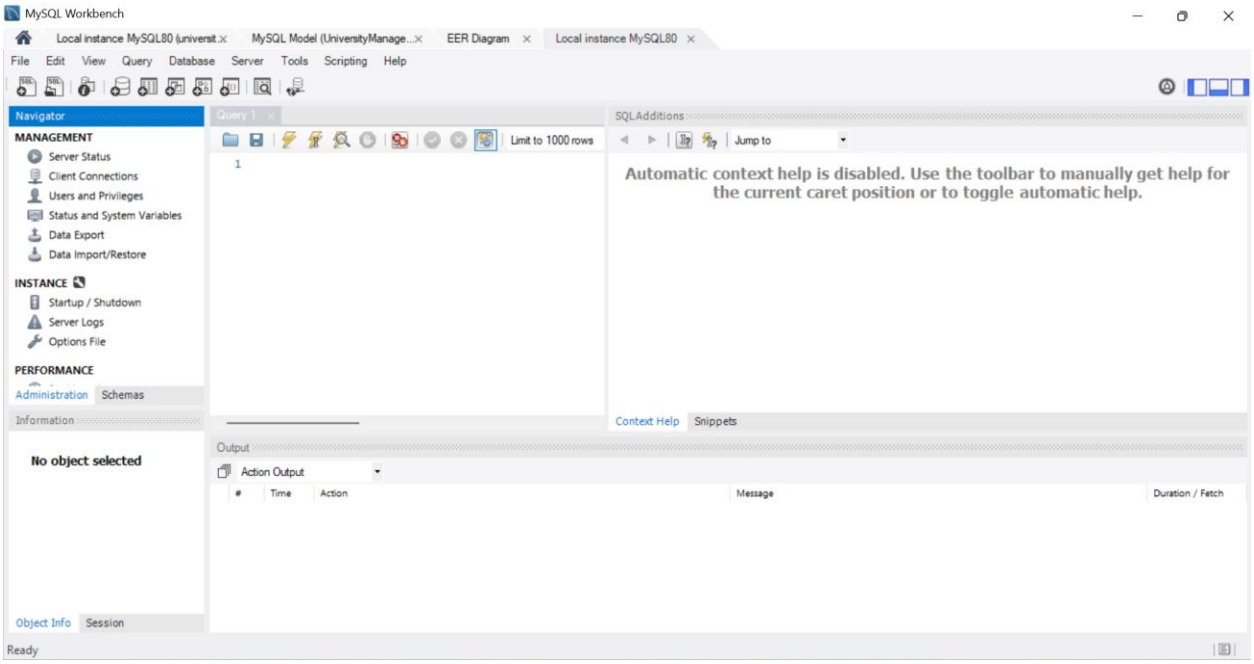
COMMIT;

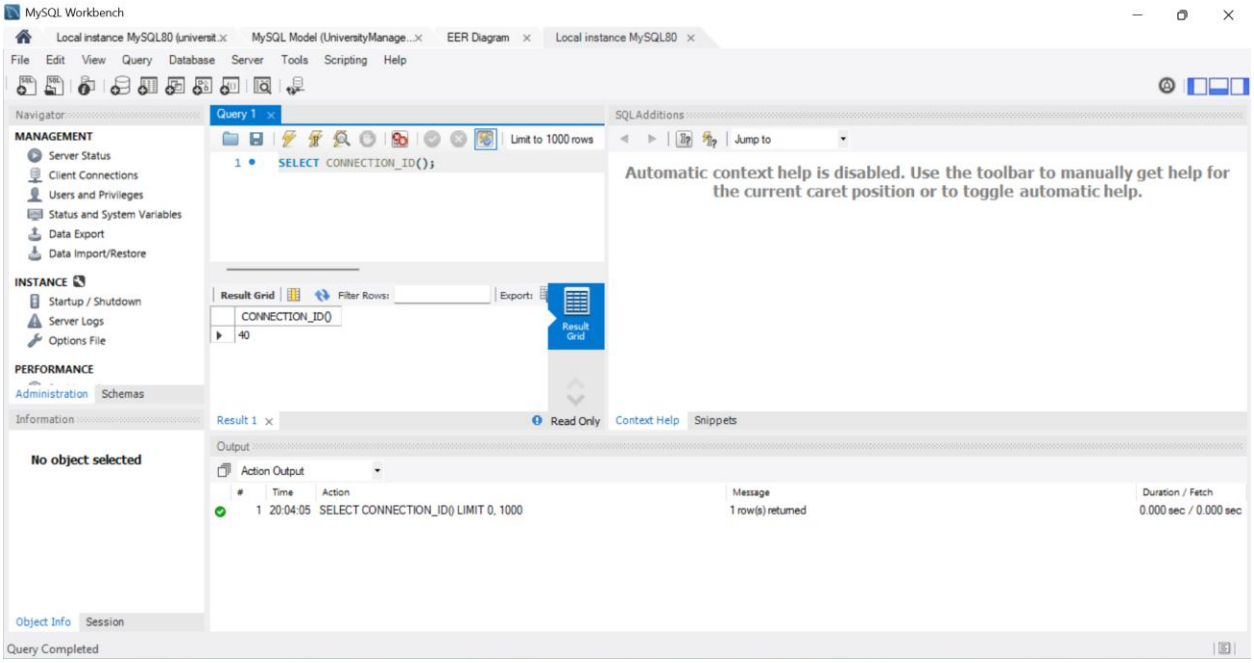


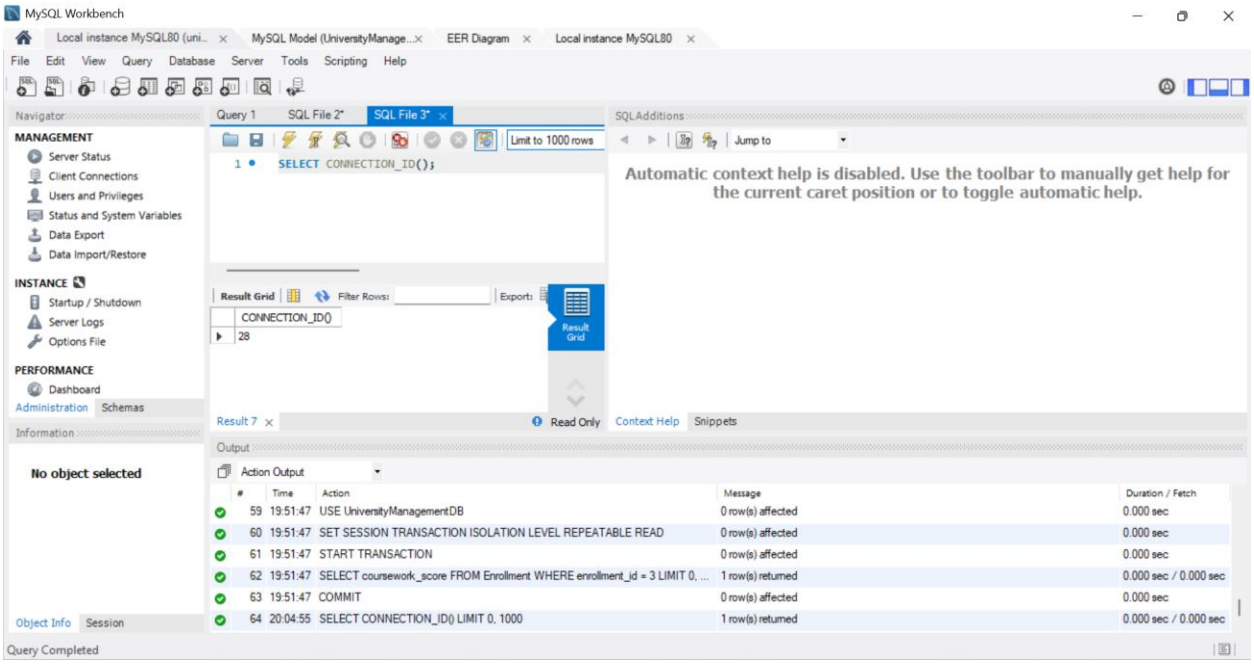
Demonstration of Concurrency using two MySQL Sessions

Screenshots showing the opening of a Second Session




MySQL Workbench
Local instance MySQL80 (universit...
MySQL Model (UniversityManage...
EER Diagram
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
MANAGEMENT
Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
INSTANCE
Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Administration Schemas
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
Ready


MySQL Workbench
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Options File
PERFORMANCE
Administration Schemas
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1 SELECT CONNECTION_ID();
Result Grid Filter Rows: Export:
CONNECTION_ID()
40
Result 1
Read Only
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
1 20:04:05 SELECT CONNECTION_ID() LIMIT 0, 1000
1 row(s) returned
0.000 sec / 0.000 sec
Query Completed

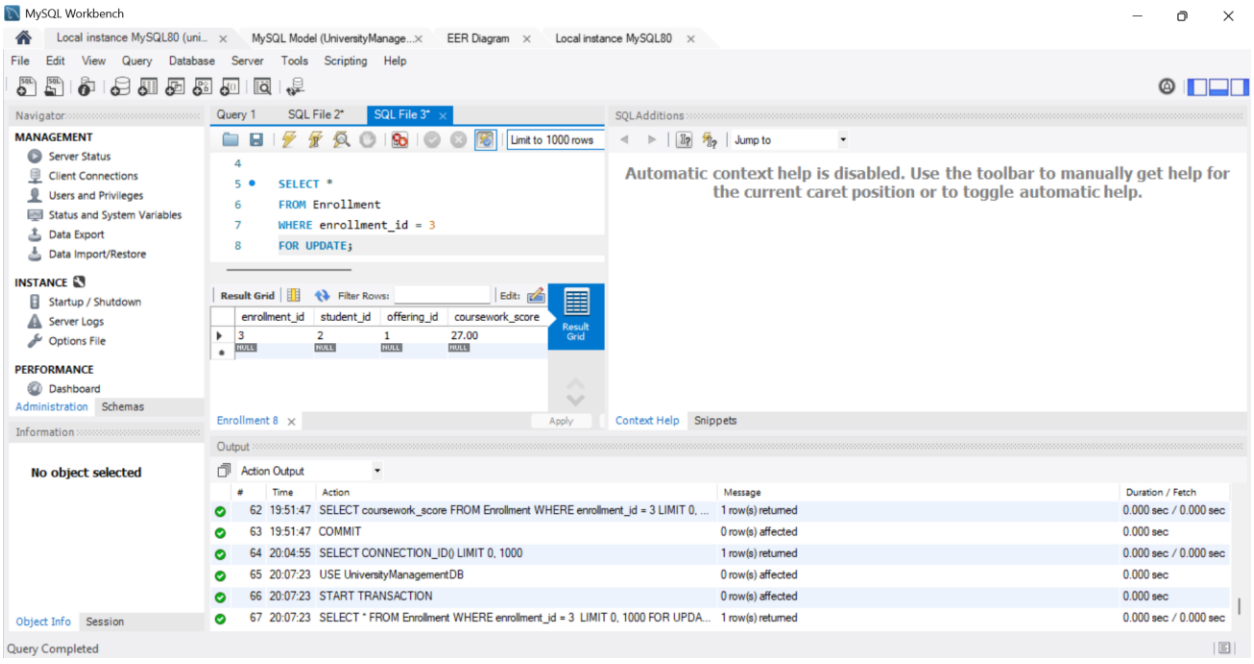


In the <u>first</u> MYSQL WorkBench, Mary's Record is to be Locked below.

```
USE UniversityManagementDB;

START TRANSACTION;

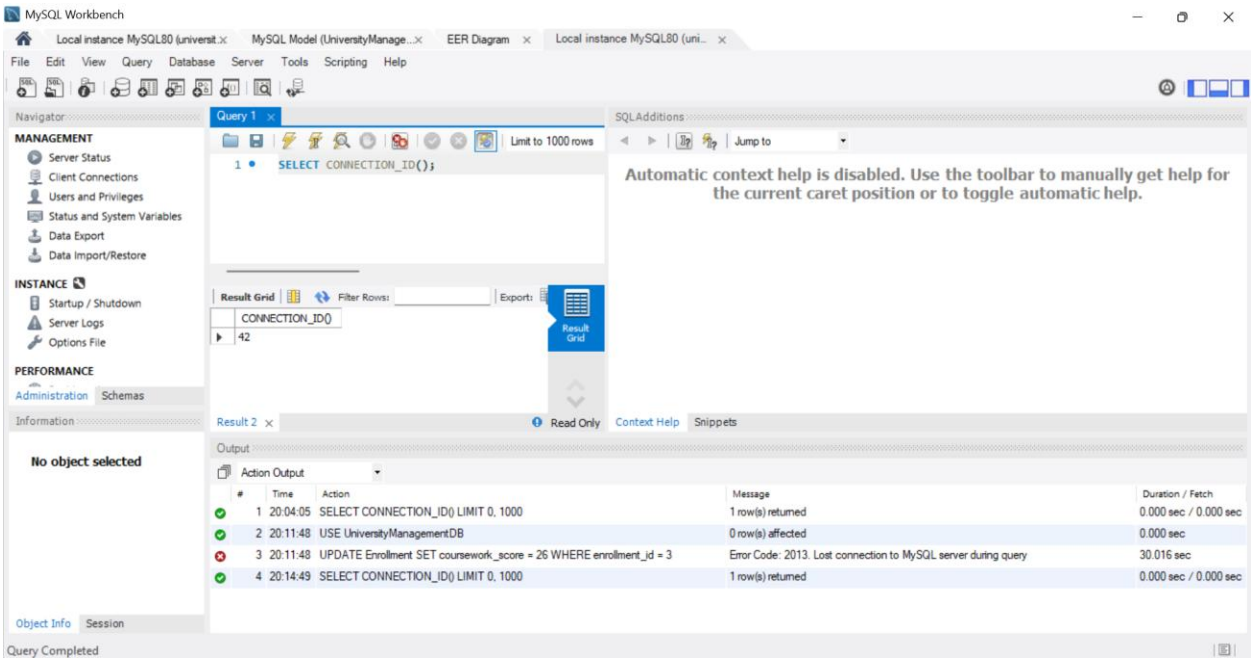
SELECT *
FROM Enrollment
WHERE enrollment_id = 3
FOR UPDATE;
```



In the Second MYSQL WorkBench an attempt to Update.

```
USE UniversityManagementDB;

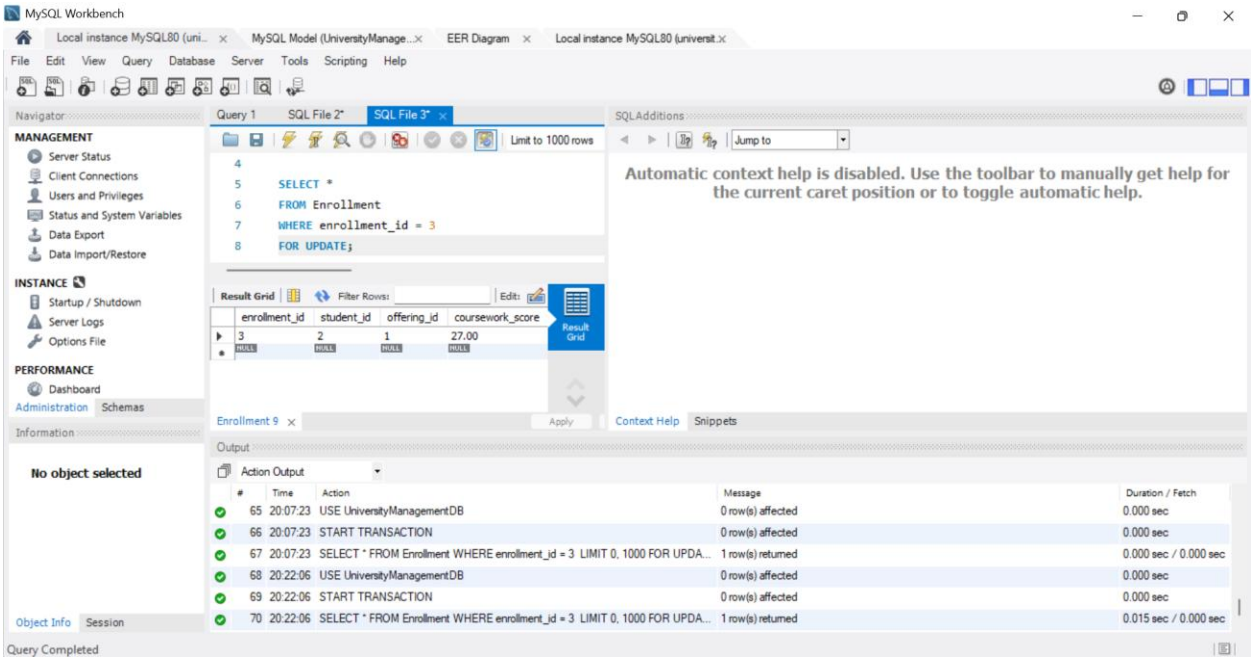
UPDATE Enrollment
SET coursework_score = 26
WHERE enrollment_id = 3;
```




```
USE UniversityManagementDB;

START TRANSACTION;

SELECT *
FROM Enrollment
WHERE enrollment_id = 3
FOR UPDATE;
```

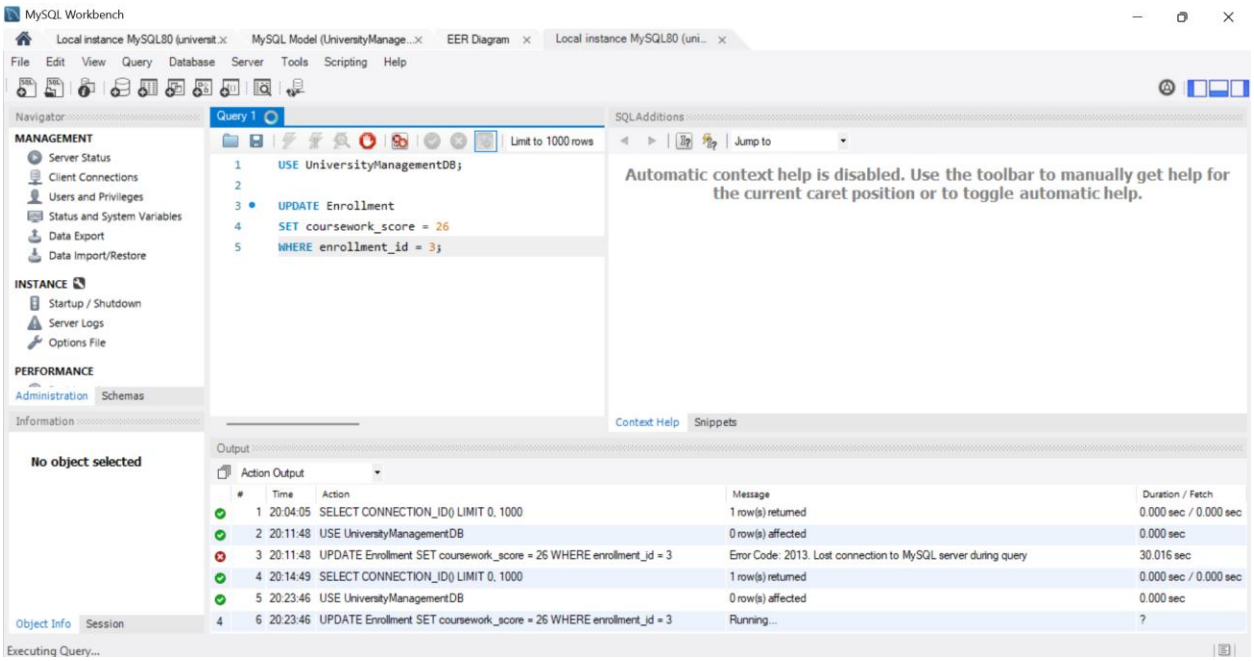


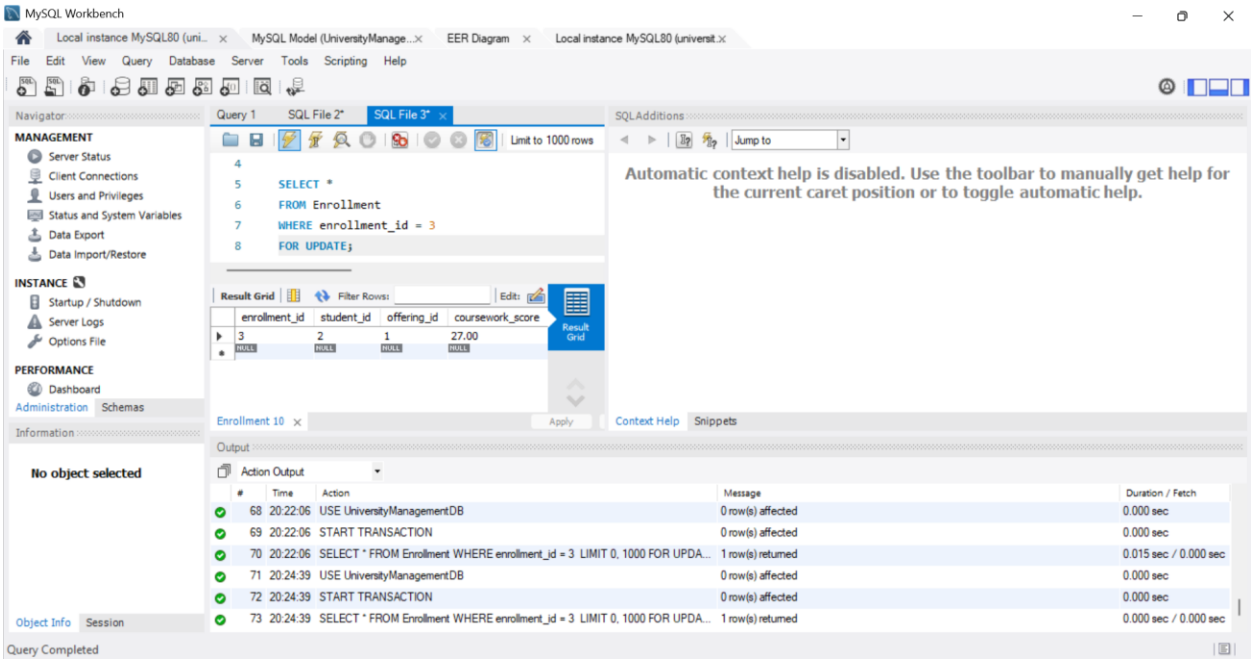
USE UniversityManagementDB;

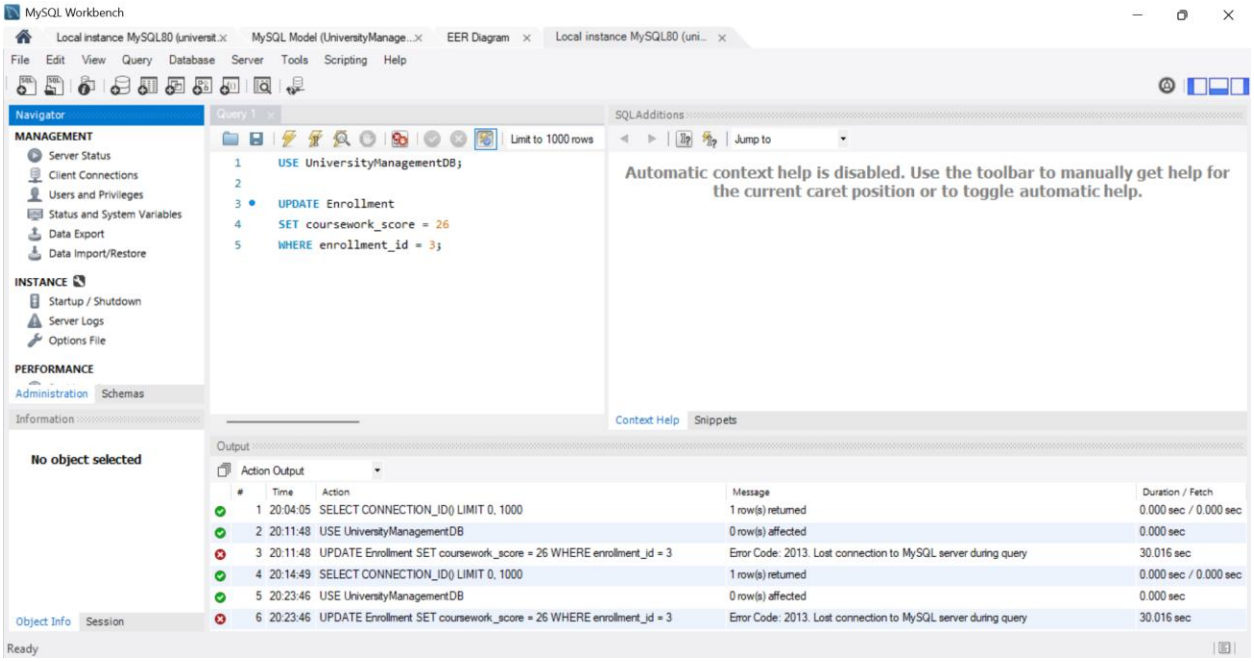
UPDATE Enrollment

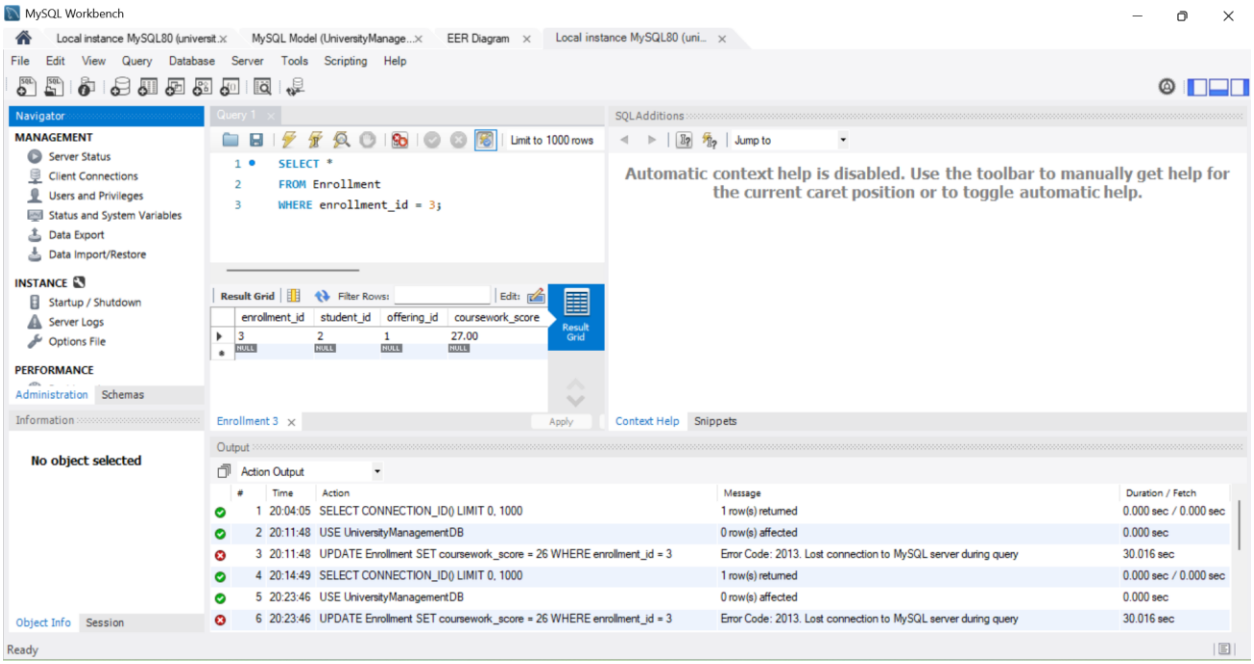
SET coursework_score = 26

WHERE enrollment_id = 3;




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Local instance MySQL80 (universit...
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Dashboard
Administration Schemas
Information
No object selected
Object Info Session
Query 1 SQL File 2* SQL File 3*
Limit to 1000 rows
4
5 SELECT *
6 FROM Enrollment
7 WHERE enrollment_id = 3
8 FOR UPDATE;
Result Grid Filter Rows: Edit:
enrollment_id student_id offering_id coursework_score
3 2 1 27.00
Result Grid
Enrollment 10
Apply
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
68 20:22:06 USE UniversityManagementDB 0 row(s) affected 0.000 sec
69 20:22:06 START TRANSACTION 0 row(s) affected 0.000 sec
70 20:22:06 SELECT * FROM Enrollment WHERE enrollment_id = 3 LIMIT 0, 1000 FOR UPDA... 1 row(s) returned 0.015 sec / 0.000 sec
71 20:24:39 USE UniversityManagementDB 0 row(s) affected 0.000 sec
72 20:24:39 START TRANSACTION 0 row(s) affected 0.000 sec
73 20:24:39 SELECT * FROM Enrollment WHERE enrollment_id = 3 LIMIT 0, 1000 FOR UPDA... 1 row(s) returned 0.000 sec / 0.000 sec
Query Completed


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Information
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Object Info Session
Query 1
Limit to 1000 rows
1 USE UniversityManagementDB;
2
3 UPDATE Enrollment
4 SET coursework_score = 26
5 WHERE enrollment_id = 3;
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
1 20:04:05 SELECT CONNECTION_ID() LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
2 20:11:48 USE UniversityManagementDB 0 row(s) affected 0.000 sec
3 20:11:48 UPDATE Enrollment SET coursework_score = 26 WHERE enrollment_id = 3 Error Code: 2013. Lost connection to MySQL server during query 30.016 sec
4 20:14:49 SELECT CONNECTION_ID() LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 20:23:46 USE UniversityManagementDB 0 row(s) affected 0.000 sec
6 20:23:46 UPDATE Enrollment SET coursework_score = 26 WHERE enrollment_id = 3 Error Code: 2013. Lost connection to MySQL server during query 30.016 sec
Ready



USE UniversityManagementDB;

UPDATE Enrollment

SET coursework_score = 26

WHERE enrollment_id = 3;

